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### Work function tuning in semiconductor devices

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#### Abstract

The present disclosure describes a method for forming a semiconductor device having a work function metal layer doped with tantalum to mitigate oxygen diffusion and improve device threshold voltage. The method includes forming a gate dielectric layer on a channel structure and forming a work function metal layer on the gate dielectric layer. The gate dielectric layer includes an interfacial layer on the channel structure and a high-k dielectric layer on the interfacial layer. The method further includes doping the work function metal layer and the gate dielectric layer with tantalum.

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## **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/324,851, titled “Tantalum Soak Method for P Work Function Tuning,” which was filed on Mar. 29, 2022 and is incorporated herein by reference in its entirety.

### **BACKGROUND**

(1) With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs, fin field effect transistors (finFETs), and gate-all-around field effect transistors (GAAFETs). Such scaling down has introduced challenges to improve the performance of semiconductor devices.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures.

(2) FIGS. 1, 2, 3A, 3B, and 4 illustrate isometric and cross-sectional views of a semiconductor device having a work function metal layer doped with tantalum, in accordance with some embodiments.

(3) FIG. 5 is a flow diagram of a method for fabricating a semiconductor device having a work function metal layer doped with tantalum, in accordance with some embodiments.

(4) FIG. 6 illustrates an isometric view of a partially-fabricated semiconductor device, in accordance with some embodiments.

(5) FIGS. 7-10, 11A-11C, 12-14, 15A-15B, and 16-20 illustrate various partial cross-sectional views of a semiconductor device having a work function metal layer doped with tantalum at various stages of its fabrication process, in accordance with some embodiments.

(6) Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

### **DETAILED DESCRIPTION**

(7) The following disclosure provides many different embodiments, or examples, for implementing, different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. As used herein, the formation of a first feature on a

second feature means the first feature is formed in direct contact with the second feature. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(8) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(9) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(10) It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

(11) In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g.,  $\pm 1\%$ ,  $\pm 2\%$ ,  $\pm 3\%$ ,  $\pm 4\%$ ,  $\pm 5\%$  of the value) These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

(12) With increasing demand for lower power consumption, higher performance, and smaller semiconductor devices, dimensions of semiconductor devices continue to scale down. The continuous scaling down of device dimensions and the increasing demand for device performance may require various process and material improvements, which can lead to various challenges. For example, n-type field effect transistors (also referred to as “NFETs”) and p-type field effect transistors (also referred to as “PFETs”) can be manufactured with different threshold voltages ( $V_{sub.t}$ ) suitable for each type of FET. The term “p-type” can be associated with a structure, layer, and/or region doped with p-type dopants, such as boron. The term “n-type” can be associated with a structure, layer, and/or region doped with n-type dopants, such as phosphorus. Devices can have different work function metal layers for different  $V_{sub.t}$ . The work function metal layers for NFET can be referred to as “n-type work function metal layers” and the work function metal layers for PFET can be referred to as “p-type work function metal layers.”

(13) The work function metal layers for NFET can include titanium aluminum (TiAl), titanium nitride (TiN), and other suitable work function materials. The work function metal layers for PFET can include titanium nitride (TiN), titanium silicon nitride (TiSiN), tungsten nitride (WN), tungsten carbon nitride (WCN), and other suitable work function materials. During the manufacturing process of p-type work function metal layers, such as TiN, subsequent processes may be in a different chamber and the surface of the p-type work function metal layers may be oxidized during the process. The oxygen at the surface can diffuse into gate dielectric layer of PFET, The gate dielectric layer can include a high-k dielectric layer and an interfacial layer. The term “high-k” can refer to a high dielectric constant. In the field of semiconductor device structures and manufacturing processes, high-k can refer to a dielectric constant that is greater than the dielectric constant of  $SiO_2$  (e.g., greater than about 3.9). The interfacial layer can include silicon oxide,

germanium oxide, or silicon germanium oxide. The diffused oxygen in the high-k dielectric layer and the interfacial layer can lead to  $V_{sub.t}$  shifts in PFETs. For example, the  $V_{sub.t}$  can increase about 20 mV to about 100 mV due to oxygen diffusion. The  $V_{sub.t}$  shift can significantly degrade the device performance of PFETs.

(14) Various embodiments of the present disclosure provide methods for forming a semiconductor device having a work function metal layer doped with tantalum to prevent oxygen diffusion into the gate dielectric layer and improve device threshold voltage. In some embodiments, a gate dielectric layer can be formed on a channel structure of a semiconductor device. The gate dielectric layer can include a high-k dielectric layer and an interfacial layer. A work function metal layer can be formed on the gate dielectric layer. In some embodiments, the work function metal layer and the gate dielectric layer can be doped with tantalum by a soak process with a tantalum precursor. The tantalum in the work function metal layer and the gate dielectric layer can attract oxygen and prevent oxygen from diffusing into the gate dielectric layer. In some embodiments, the work function metal layer can include one or more tantalum nitride layers to dope tantalum in the work function metal layer and the gate dielectric layer and thus to mitigate oxygen diffusion. In some embodiments, the work function metal layer can include titanium tantalum nitride (TiTa<sub>N</sub>) to mitigate oxygen diffusion. The tantalum in the TiTa<sub>N</sub> work function metal layer can have a concentration ranging from about 0.05% to about 25%, and the tantalum can diffuse into the gate dielectric layer to attract oxygen and mitigate oxygen diffusion. In some embodiments, the tantalum concentration in the work function metal layer and the gate dielectric layer can range from about 0.05% to about 25%. In some embodiments, with tantalum doping in the work function metal layer and the gate dielectric layer,  $V_{sub.t}$  shifts in PFETs can be reduced by about 20 mV to about 100 mV and device performance of the semiconductor device can be improved.

(15) FIGS. 1, 2, 3A, 3B, and 4 illustrate isometric and cross-sectional -views of a semiconductor device **100** having a work function layer doped with tantalum, in accordance with some embodiments. As shown in FIG. 1, semiconductor device **100** includes a FET **105A** and a FET **105B** formed on a substrate **125**. In some embodiments, FETs **105A** and **105B** can be finFETs, planar FETs, nanostructure transistors, or other suitable FET devices. The nanostructure transistors can include nanosheet transistors, nanowire transistors, multi bridge channel transistor, nano-ribbon transistor, etc. The nanostructure transistors provide a channel in a stacked nanosheet/nanowire configuration. Though FIGS. 1, 2, 3A, 3B, and 4 illustrate FETs **105A** and **105B** as nanostructure transistors, FETs **105A** and **105B** can be any suitable FET devices. In some embodiments, FETs **105A** and **105B** can be both PFETs, both NFETs, or one of each conductivity type FET. Though FIG. 1 shows two FETs, semiconductor device **100** can have any number of FETs similar to FETs **105A** and **105B**. Also, though FIG. 1 shows one gate structure **155**, semiconductor device **100** can have additional gate structures similar and parallel to gate structure **155**. The discussion of elements of FETs **105A** and **105B** with the same annotations applies to each other, unless mentioned otherwise.

(16) As shown in FIG. 1, FETs **105A** and **105B** can be formed on substrate **125**. In some embodiments, substrate **125** can include a semiconductor material, such as crystalline silicon (Si). In some embodiments, substrate **125** can include (i) an elementary semiconductor, such as germanium (Ge); (ii) a compound semiconductor including silicon carbide (SiC), silicon arsenide (SiAs), gallium arsenide (GaAs), gallium phosphide (GaP), and/or a III-V semiconductor material; (iii) an alloy semiconductor including silicon germanium (SiGe), silicon germanium carbide (SiGeC), germanium stannum (GeSn), and/or aluminum gallium arsenide (AlGaAs); (iv) a silicon-on-insulator (SOI) structure; (v) a silicon germanium (SiGe)-on insulator (SiGeOI) structure; (vi) germanium-on-insulator (GeOI) structure; or (vii) a combination thereof. Alternatively, the substrate can be made from an electrically non-conductive material, such as glass and sapphire wafer. Further, substrate **125** can be doped depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, substrate **125** can be doped with p-type dopants (e.g.,

boron, indium, aluminum, or gallium) or n-type dopants (e.g., phosphorus or arsenic). For example purposes, substrate **125** will be described in the context of crystalline Si. Based on the disclosure herein, other materials, as discussed above, can be used. These materials are within the spirit and scope of this disclosure.

(17) Referring to FIG. **1**, semiconductor device **100** can include additional structural elements, such as fin structures **120**, a liner **130**, an insulating layer **135**, source/drain (S/D) structures **140**, an etch stop layer **145**, an isolation layer **150**, a gate structure **155** formed in isolation layer **150**, and gate spacers **160** formed on sidewall surfaces of gate structure **155**.

(18) Fin structures **120** may be formed on substrate **125** by patterning with any suitable method. For example, fin structures **120** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Double-patterning or multi-patterning processes can combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. In some embodiments, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern fin structures **120**. In some embodiments, fin structures **120** can include semiconductor materials similar to substrate **125**. In some embodiments, fin structures **120** can include crystalline Si. In some embodiments, fin structures **120** are optional.

(19) In some embodiments, insulating layer **135** can be an isolation structure, such as a shallow trench isolation (STI), that provides electrical isolation between FETs **105A** and **105B** from each other and from neighboring FETs (not shown) on substrate **125** and/or neighboring active and passive elements (not shown) integrated with or deposited on substrate **125**. In some embodiments, an insulating layer can be a layer that functions as an electrical insulator (e.g., a dielectric layer). In some embodiments, insulating layer **135** can include silicon oxide (SiO<sub>2</sub>), silicon nitride (Si<sub>3</sub>N<sub>4</sub>), silicon oxynitride (SiON), fluorine-doped silicate glass (FSG), phosphorous-doped silicate glass (PSG), a low-k dielectric material (e.g., with k-value less than about 3.9), and/or other suitable dielectric materials with appropriate fill properties. In some embodiments, liner **130** is a nitride layer, such as silicon nitride.

(20) Referring to FIG. **1**, S/D structures **140** can be disposed on fin structures **120** and abut gate spacers **160**, extending along an X-axis within isolation layer **150**. In some embodiments, S/D structures **140** can have any geometric shape, such as a polygon, an ellipse, and a circle. S/D structures **140** can include an epitaxially-grown semiconductor material. In some embodiments, the epitaxially-grown semiconductor material includes the same material as substrate **125**. In some embodiments, the epitaxially-grown semiconductor material includes a different material from substrate **125**. In some embodiments, the epitaxially-grown semiconductor material for each of S/D structures **140** can be the same as or different from each other. The epitaxially-grown semiconductor material can include: (i) a semiconductor material, such as germanium and silicon; (ii) a compound semiconductor material, such as gallium arsenide and aluminum gallium arsenide; or (iii) a semiconductor alloy, such as silicon germanium and gallium arsenide phosphide.

(21) In some embodiments, S/D structures **140** can be p-type for a PFET and n-type for an NFET. In some embodiments, p-type S/D structures **140** can include SiGe and can be in-situ doped during an epitaxial growth process using p-type dopants, such as boron, indium, and gallium. In some embodiments, p-type S/D structures **140** can have multiple sub-regions that can include SiGe and can differ from each other based on, for example, doping concentrations, epitaxial growth process conditions, and/or a relative concentration of Ge with respect to Si. In some embodiments, n-type S/D structures **140** can include Si and can be in-situ doped during an epitaxial growth process using n-type dopants, such as phosphorus and arsenic. In some embodiments, n-type S/D structures **140** can have multiple n-type epitaxial fin sub-regions that can differ from each other based on, for example, doping concentration and/or epitaxial growth process conditions,

(22) Referring to FIG. 1, etch stop layer **145** can extend over insulating layer **135**, S/D structures **140**, and gate spacers **160**. In some embodiments, etch stop layer **145** can function as a layer to stop etching in a subsequent etching process during the formation of S/D contact openings on S/D structures **140**. In some embodiments, etch stop layer **145** can be deposited by a conformal deposition process, such as atomic layer deposition (ALD), plasma-enhanced ALD (PEALD), chemical vapor deposition (CVD), plasma-enhanced CVD (PECVD), and any other suitable deposition method.

(23) Isolation layer **150** can surround S/D structures **140** and can be formed prior to the formation of gate structure **155**. In some embodiments, isolation layer **150** can be an interlayer dielectric (ILD) that includes a silicon oxide-based dielectric material with or without carbon and/or nitrogen. In some embodiments, isolation layer **150** can be deposited by CVD, flowable CVD (FCVD), or any other suitable deposition method.

(24) Gate spacers **160** can be a stack of one or more layers that include the same or different materials. In some embodiments, gate spacers **160** can include a dielectric material, such as silicon oxynitride (SiON), silicon carbonitride (SiCN), silicon oxycarbide (SiOC), silicon nitride, or a combination thereof. According to some embodiments, gate spacers **160** can be disposed on sidewall surfaces of gate structure **155**. Gate spacers **160** can have a low-k material with a dielectric constant less than about 3.9 (e.g., about 3.5, about 3.0, or about 2.8).

(25) Gate structure **155** can be multi-layered structures and can be disposed above fin structures **120**. Gate structure **155** can include a gate dielectric layer **203** and a metal gate **215**, as shown in detail in FIGS. 2, 3A, 3B, and 4, FIG. 2 illustrates a cross-sectional view of semiconductor device **100** across line A-A in FIG. 1, in accordance with some embodiments. FIG. 3A illustrates a top-down view of semiconductor device **100** across line C-C in FIG. 2, in accordance with some embodiments. FIG. 3B illustrates a top-down view of semiconductor device **100** across line D-D in FIG. 2, in accordance with some embodiments. FIG. 4 illustrates a cross-sectional view of semiconductor device **100** across line B-B in FIG. 1, in accordance with some embodiments.

(26) As shown in FIGS. 2, 3A, 3B, and 4, gate dielectric layer **203** can include an interfacial layer **207** and a high-k dielectric layer **209**. In some embodiments, interfacial layer **207** can include silicon oxide, germanium oxide, or silicon germanium oxide with a thickness from about 5 Å to about 15 Å. In some embodiments, high-k dielectric layer **209** can include a dielectric material with a dielectric constant (k-value) higher than about 3.9. In some embodiments, high-k dielectric layer **209** can include hafnium oxide, aluminum oxide, zirconium oxide, or other suitable high-k dielectric materials deposited by ALD, CVD, or PEALD with a thickness from about 10 Å to about 75 Å.

(27) In some embodiments, gate dielectric layer **203** can be doped with tantalum to mitigate oxygen diffusion to interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. In some embodiments, the concentration of the doped tantalum in interfacial layer **207** and high-k dielectric layer **209** can range from about 0.05% to about 25%. If the tantalum concentration is less than about 0.05%, the doped tantalum may not be able to prevent oxygen from diffusing to interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. As a result, the electrical performance of semiconductor device **100** may not be improved. If the tantalum concentration is greater than about 25%, the excessive tantalum may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**. The decrease of the uniformity of gate dielectric layer **203** can degrade the electrical performance of semiconductor device **100**.

(28) In some embodiments, metal gate **215** can include a work function metal layer **211** and a metal fill **213**. Work function metal layer **211** can include work function metals to tune  $V_{sub.t}$  of FETs **105A** and **105B**. In some embodiments, work function layer **211** can include p-type work function metals, such as TiN, TiSiN, WN, WCN, or other suitable work function metals. In some embodiments, work function layer **211** can include a single metal layer or a stack of metal layers.

The stack of metal layers can include work function metals having work function values equal to or different from each other. In some embodiments, as shown in FIG. 2, work function metal layer **211** can have a thickness **211t** ranging from about 1 nm to about 10 nm. If thickness **211t** is less than about 1 nm,  $V_{sub.t}$  of FETs **105A** and **105B** may be less than the required value (e.g., from about 50 mV to about 500 mV). If thickness **211t** is greater than about 10 nm,  $V_{sub.t}$  of FETs **105A** and **105B** may be greater than required value (e.g., from about 50 mV to about 500 mV).

(29) In some embodiments, work function metal layer **211** can be doped with tantalum to mitigate oxygen diffusion to interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. In some embodiments, the concentration of the doped tantalum in work function metal layer **211** can range from about 0.05% to about 25%. If the tantalum concentration is less than about 0.05%, the doped tantalum may not be able to prevent oxygen from diffusing to interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. As a result, the electrical performance of semiconductor device **100** may not be improved. If the tantalum concentration is greater than about 25%, the excessive tantalum may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**. The decrease of the uniformity of gate dielectric layer **203** can degrade the electrical performance of semiconductor device **100**.

(30) In some embodiments, metal fill **213** can include titanium, tantalum, aluminum, cobalt, tungsten, nickel, ruthenium, or other suitable conductive materials. In some embodiments, interfacial layer **207** and a high-k dielectric layer **209** can wrap around nanostructures **210**. One or more layers of work function metal layer **211** and metal fill **213** can fill the spaces between nanostructures **210**. Accordingly, gate structure **155** can be referred to as “gate-all-around (GAA) structures” and FETs **105A** and **105B** can be referred to as “GAA FETs.”

(31) As shown in FIGS. 2, 3A, 3B, and 4, semiconductor device can further include nanostructures **210** and inner spacer structures **217**. In some embodiments, nanostructures **210** can include a stack of semiconductor layers (e.g., a stack of nanosheets, nanowires, nanoribbons, or nano-fork sheets for GAA FETs). In some embodiments, nanostructures **210** can include semiconductor materials similar to or different from fin structures **120**. In some embodiments, nanostructures **210** and fin structures **120** can include a semiconductor material the same as substrate **125**, such as crystalline Si. In some embodiments, nanostructures **210** can include silicon germanium. In some embodiments, as shown in FIG. 2, nanostructures **210** can have a thickness **210t** along a Z-axis ranging from about 5 nm to about 10 nm. In some embodiments, adjacent nanostructures **210** can have a spacing **210s** along a Z-axis ranging from about 5 nm to about 15 nm.

(32) In some embodiments, inner spacer structures **217** can isolate gate structures **155** and S/D structures **140**. Inner spacer structures **217** can include insulating materials, such as silicon oxide, silicon nitride, SiON, SiCN, SiOC, silicon oxycarbonitride (SiOCN), a low-k material, and a combination thereof. In some embodiments, inner spacer structures **217** and gate spacers **160** can include the same insulating material. In some embodiments, inner spacer structures **217** and gate spacers **160** can include different insulating materials. Inner spacer structures **217** can include a single layer or a stack of insulating layers. In some embodiments, inner spacer structures **217** can have a low-k material with a dielectric constant less than about 3.9 (e.g., about 3.5, about 3.0, or about 2.8). In some embodiments, inner spacer structures **217** can have a thickness **217t** along an X-axis ranging from about 4 nm to about 8 nm.

(33) Referring to FIGS. 1, 2, 3A, 3B, and 4, nanostructures **210** can be current-carrying structures for respective FETs **105A** and **105B**. Channel regions of FETs **105A** and **105B** can be formed in portions of their respective nanostructures **210**. As a result, nanostructures **210** can be referred to as “channel structures.” S/D structures **140** can function as S/D regions of respective FETs **105A** and **105B**.

(34) FIG. 5 is a flow diagram of method **500** for fabricating a semiconductor device having a work function metal layer doped with tantalum, according to some embodiments. Method **500** may not

be limited to nanostructure devices and can be applicable to other devices that would benefit from tantalum doped work function metal layer, such as planar FETs, finFETs, etc. Additional fabrication operations may be performed between various operations of method **500** and may be omitted merely for clarity and ease of description. Additional processes can be provided before, during, and/or after method **500**; one or more of these additional processes are briefly described herein. Moreover, not all operations may be needed to perform the disclosure provided herein. Additionally, some of the operations may be performed simultaneously or in a different order than shown in FIG. **5**. In some embodiments, one or more other operations may be performed in addition to or in place of the presently described operations.

(35) For illustrative purposes, the operations illustrated in FIG. **5** will be described with reference to the example fabrication process for fabricating semiconductor device **100** having a tantalum doped work function metal layer as illustrated in FIGS. **6-10**, **11A-11C**, **12-14**, **15A-15B**, and **16-20**. FIG. **6** illustrates an isometric view of a partially-fabricated semiconductor device **100**, in accordance with some embodiments. FIGS. **7-10**, **11A-11C**, **12-14**, **15A-15B**, and **16-20** illustrate partial cross-sectional views of semiconductor device **100** having tantalum doped work function metal layer **211** at various stages of its fabrication process, in accordance with some embodiments. One of FET **105A** or FET **105B** in semiconductor device **100** is illustrated in FIGS. **7-10**, **11A-11C**, **12-14**, **15A-15B**, and **16-20** for simplicity. Although FIGS. **6-10**, **11A-11C**, **12-14**, **15A-15B**, and **16-20** illustrate tantalum doping in work function metal layer **211** and gate dielectric layer **203** for semiconductor device **100**, method **500** can be applied to other semiconductor devices, such as planar FETs, finFETs, and other suitable devices. Elements in FIGS. **6-10**, **11A-11C**, **12-14**, **15A-15B**, and **16-20** with the same annotations as elements in FIGS. **1**, **2**, **3A**, **3B**, and **4** are described above.

(36) Referring to FIG. **5**, method **500** begins with operation **510** and the process of forming a gate dielectric layer on a channel structure. For example, as shown in FIGS. **6-8**, gate dielectric layer **203** can be formed on nanostructures **210**. According to some embodiments, FIG. **6** illustrates an isometric views of partially-fabricated semiconductor device **100** after the removal of a sacrificial gate stack. After the removal of the sacrificial gate stack, a gate stack opening **655** can be formed between gate spacers **160**. FIG. **7** illustrates a partial cross-sectional view of semiconductor device **100** across line E-E in FIG. **6**, in accordance with some embodiments. FIG. **8** is a cross-sectional view of semiconductor device **100** shown in FIG. **7** after operation **510** of method **500**. As shown in FIGS. **6** and **7**, after the removal of the sacrificial gate stack, nanostructures **210** can be formed in gate stack opening **655** between S/D structures **140**. In some embodiments, as shown in FIG. **7**, nanostructures **210** can have thickness  $210t$  along a Z-axis ranging from about 5 nm to about 110 nm. In some embodiments, adjacent nanostructures **210** can have spacing  $210s$  along a Z-axis ranging from about 5 nm to about 15 nm. Nanostructures **210** can be current-carrying structures for FETs **105A** and **105B** and can form the channel regions of FETs **105A** and **105B**. As a result, nanostructures **210** can be referred to as “channel structures” of FETs **105A** and **105B**.

(37) As shown in FIG. **8**, gate dielectric layer **203** can be formed on nanostructures **210**, fin structures **120**, and insulating layer **135**. Gate dielectric layer **203** can be formed in gate stack opening **655** between gate spacers **160**, as shown in FIG. **6**. In some embodiments, gate dielectric layer **203** can include interfacial layer **207** and high-k dielectric layer **209**. In some embodiments, interfacial layer **207** can be formed by exposing the silicon surfaces of nanostructures **210** and fin structures **120** to an oxidizing ambient. In some embodiments, the oxidizing ambient can include a combination of ozone ( $O_{sub.3}$ ), ammonia hydroxide/hydrogen peroxide/water mixture (SC1), and hydrochloric acid/hydrogen peroxide/water mixture (SC2). As a result of the aforementioned oxidation process, a silicon oxide layer between about 5 Å and about 15 Å can be formed on exposed silicon surfaces, such as the surfaces of nanostructures **210** and fin structures **120** in gate stack opening **655**, but not on insulating layer **135**. Therefore, gate dielectric layer **203** on nanostructures **210** and fin structures **120** can include interfacial layer **207** and high-k dielectric

layer **209**, and gate dielectric layer **203** on insulating layer **1135** can include only high-k dielectric layer **209**, according to some embodiments. In some embodiments, interfacial layer **207** can include a silicon oxide layer with a thickness from about 5 Å to about 15 Å and deposited by ALD, CVD, or any other suitable deposition method. As a result of the deposition process, the silicon oxide layer can cover nanostructures **210**, fin structures **120**, and insulating layer **135**. In some embodiments, high-k dielectric layer **209** can include a dielectric material with a dielectric constant (k-value) higher than about 3.9. In some embodiments, high-k dielectric layer **209** can include hafnium oxide, aluminum oxide, zirconium oxide, or other suitable high-k dielectric materials deposited by ALD, CVD, or PEALD with a thickness from about 10 Å to about 75 Å.

(38) Referring to FIG. 5, method **500** continues with operation **520** and the process of forming a work function metal layer on the gate dielectric layer. For example, as shown in FIG. 9, work function metal layer **211\*** can be formed on gate dielectric layer **203**. FIG. 9 illustrates a cross-sectional view of semiconductor device **100** shown in FIG. 8 after operation **520**, in accordance with some embodiments. In some embodiments, work function layer **211\*** can include TiN, TiSiN, WN, WCN, or other suitable work function metals. In some embodiments, work function layer **211\*** can be deposited on gate dielectric layer **203** by ALD, CVD, and other suitable deposition methods at a temperature from about 150° C. to about 550° C. under a pressure from about 0.1 torr to about 50 torr. In some embodiments, work function metal layer **211\*** can have a thickness ranging from about 1 nm to about 10 nm.

(39) Referring to FIG. 5, in operation **530**, the work function metal layer and the gate dielectric layer is doped with tantalum. For example, as shown in FIGS. 10 and 11A-11C, work function metal layer **211\*** and gate dielectric layer **203** can be doped with tantalum by a soak process. FIG. 10 illustrates a cross-sectional view of semiconductor device **100** shown in FIG. 9 after operation **530**, in accordance with some embodiments. FIGS. 11A-11C illustrate enlarged cross-sectional views of region **1070** in FIG. 10, in accordance with some embodiments. In some embodiments, the arrows in FIGS. 10 and 11A-11C can indicate the tantalum diffusion during the soak process. In some embodiments, the soak process can use a tantalum precursor, such as PDMAT ( $\text{Ta}(\text{N}(\text{CH}_3)_3)_2\text{Cl}_5$ ) gas and tantalum chloride ( $\text{TaCl}_5$ ) gas. In some embodiments, the soak process can be performed at a temperature from about 150° C. to about 550° C. under a pressure from about 0.1 torr to about 50 torr. If the temperature is less than about 150° C., or the pressure is less than about 0.1 torr, tantalum may not be doped in work function metal layer **211\*** or gate dielectric layer **203**. As a result, oxygen diffused into gate dielectric layer **203** may not be mitigated and the electrical performance of semiconductor device **100** may not be improved. If the temperature is greater than about 150° C., or the pressure is greater than about 50 torr, excessive tantalum may be doped in work function metal layer **211\*** and gate dielectric layer **203**, and the excessive tantalum may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**. The decrease of the uniformity of gate dielectric layer **203** can degrade the electrical performance of semiconductor device **100**.

(40) In some embodiments, the soak process can be performed for a time period ranging from about 0.5 s to about 1800 s. The time period of the soak process can control the tantalum concentration in work function metal layer **211\*** and gate dielectric layer **203**. If the time period is less than about 0.5 s, the tantalum concentration in work function metal layer **211\*** and gate dielectric layer **203** may be less than about 0.05% and oxygen diffused into gate dielectric layer **203** may not be mitigated. If the time period is greater than about 1800 s, the tantalum concentration in work function metal layer **211\*** and gate dielectric layer **203** may be greater than about 25% and the excessive tantalum in gate dielectric layer **203** may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**.

(41) In some embodiments, as shown in FIG. 11A, the soak process can be performed on high-k dielectric layer **209** before the deposition of work function metal layer **211\***. The tantalum can diffuse into gate dielectric layer **203** and mitigate oxygen diffusion to interfacial layer **207** and the

interface between interfacial layer **207** and high-k dielectric layer **209**. In some embodiments, as shown in FIG. **11B**, the soak process can be performed after the deposition of work function metal layer **211\***. The tantalum can diffuse through work function metal layer **211\*** and dope work function metal layer **211\*** and gate dielectric layer **203**. In some embodiments, as shown in FIG. **11C**, the soak process can be performed multiple times after the deposition of each work function metal sublayers **211-1\***, **211-2\***, and **211-3\***. In some embodiments, each of work function metal sublayers **211-1\***, **211-2\***, and **211-3\*** can have a thickness ranging from about 0.5 nm to about 1.5 nm.

(42) Referring to FIG. **5**, in operation **540**, a metal fill can be formed on the work function metal layer. For example, as shown in FIGS. **12** and **13**, metal fill **213** can be formed on work function metal layer **211**. In some embodiments, prior to the formation of metal fill **213**, a glue layer can be deposited on work function metal layer **211\***, as shown in FIG. **12**. The glue layer can include the same conductive material as work function metal layer **211**, such as TiN. Accordingly, the glue layer can be part of work function metal layer **211**, as shown in FIG. **12**. In some embodiments, the glue layer can be deposited by ALD, CVD, and other suitable deposition methods at a temperature from about 150° C. to about 550° C. under a pressure from about 0.1 torr to about 50 torr. In some embodiments, the glue layer can have a thickness ranging from about 25 Å to about 200 Å.

(43) In some embodiments, metal fill **213** can include a single metal layer or a stack of metal layers. The stack of metal layers can include metals different from each other. In some embodiments, metal fill **213** can include a suitable conductive material, such as titanium, tantalum, aluminum, cobalt, tungsten, nickel, ruthenium, metal alloys, and/or combinations thereof. In some embodiments, metal fill **213** can be deposited by CVD, physical vapor deposition (PVD), and other suitable deposition methods. After the deposition of metal fill **213**, work function metal layer **211** and metal fill **213** can form metal gate **215**.

(44) In some embodiments, one or more tantalum nitride layers can be deposited on gate dielectric layer **203** during the formation of work function metal layer **211** to dope tantalum in work function metal layer **211** and gate dielectric layer **203**, as shown in FIGS. **14**, **15A**, **15B**, **16**, and **17**. FIG. **14** illustrates a cross-sectional view of semiconductor device **100** shown in FIG. **8** after operations **520** and **530**, in accordance with some embodiments. FIGS. **15A** and **15B** illustrate enlarged cross-sectional views of region **1470** in FIG. **14**, in accordance with some embodiments. In some embodiments, as shown in FIG. **15A**, work function metal layer **211\*** can include work function metal sublayers **211A1**, **211A2**, and **211A3**. Work function metal sublayers **211A1** and **211A3** can include the same work function material, such as TiN. Work function metal sublayer **211A2** can include another work function material, such as TaN, different from work function metal sublayers **211A1** and **211A3**. In some embodiments, as shown in FIG. **15B**, work function metal layer **211\*** can include work function metal sublayers **211B1**, **211B2**, **211B3**, and **211B4**. Work function metal sublayers **211B1** and **211B3** can include the same work function material, such as TiN. Work function metal sublayer **211B2** and **211B4** can include another work function material, such as TaN, different from work function metal sublayers **211B1** and **211B3**.

(45) In some embodiments, work function metal sublayers **211A1**, **211A2**, and **211A3** can be deposited on gate dielectric layer **203** by ALD, CVD, and other suitable deposition methods at a temperature from about 150° C. to about 550° C. under a pressure from about 0.1 torr to about 50 torr. In some embodiments, work function metal sublayers **211A1** and **211A3** can be deposited with a titanium precursor in the same chamber. The titanium precursor can include titanium chloride (TiCl<sub>4</sub>) or other suitable titanium precursor. In some embodiments, work function metal sublayer **211A2** can be deposited with a tantalum precursor in another chamber different from work function metal sublayers **211A1** and **211A3**. The tantalum precursor can include titanium PDMAT, TaCl<sub>5</sub>, or other suitable tantalum precursor. Accordingly, after deposition of work function metal sublayer **211A1**, the deposition process for work function metal sublayer **211A2** can have a vacuum break, which can be referred to as an “ex situ” deposition. Similarly, the deposition process for work

function metal sublayer **211A3** after deposition of work function metal sublayer **211A2** can be an ex situ deposition. In some embodiments, work function metal sublayers **211A1**, **211A2**, and **211A3** can be deposited in the same chamber with a titanium precursor for work function metal sublayers **211A1** and **211A3** and a tantalum precursor for work function metal sublayer **211A2**. The deposition process of work function metal sublayers **211A1**, **211A2**, and **211A3** can have no vacuum break, which can be referred to as an “in situ” deposition. Accordingly, work function metal sublayers **211A2** and **211A3** can be in situ deposited or ex situ deposited. In some embodiments, work function metal sublayers **211B1**, **211B2**, **211B3**, and **211B4** can be deposited by the same method as work function metal sublayers **211A1**, **211A2**, and **211A3**.

(46) In some embodiments, as shown in FIGS. **15A** and **15B**, work function metal sublayers **211\*** can include TiN and TaN layers stacked in an alternate configuration. In some embodiments, work function metal sublayers **211A1** and **211B1** can include TiN but may not include TaN, because TaN on gate dielectric layer **203** may increase  $V_t$  of PFET devices and degrade device performance. In some embodiments, work function metal sublayers **211A3** and **211B4** can include TiN or TaN as a top layer of work function metal layer **211\***. In some embodiments, the atomic bond of tantalum in work function metal sublayers **211A2**, **211B2**, and **211B4** can be broken during the deposition process and tantalum can diffuse to adjacent work function metal sublayers and gate dielectric layer **203** under the deposition temperature from about 150° C. to about 550° C. As a result, work function metal layers **211\*** and gate dielectric layer **203** can be doped with tantalum by the alternate configuration of stacked TiN and TaN layers after the deposition process.

(47) In some embodiments, the tantalum concentration in work function metal sublayers **211A2**, **211B2**, and **211B4** can range from about 40% to about 60%. In some embodiments, the concentration of the doped tantalum in work function metal sublayers **211A1**, **211A3**, **211B1**, and **211B3**, high-k dielectric layer **209**, and interfacial layer **207** can range from about 0.05% to about 25%. If the tantalum concentration is less than about 0.05%, the doped tantalum in gate dielectric layer **203** and work function metal sublayers **211A1**, **211A3**, **211B1**, and **211B3** may not be able to prevent oxygen from diffusing to interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. As a result, the electrical performance of semiconductor device **100** may not be improved. If the tantalum concentration is greater than about 25%, the excessive tantalum may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**. The decrease of the uniformity of gate dielectric layer **203** can degrade the electrical performance of semiconductor device **100**.

(48) In some embodiments, each of work function metal sublayers **211A1**, **211A3**, **211B1**, and **211B3** can include TiN having a thickness **211t1** ranging from about 0.5 nm to about 1.5 nm. In some embodiments, work function metal sublayers **211A2**, **211B2**, and **211B4** can include TaN having a thickness **211t2** ranging from about 0.5 nm to about 1.5 nm. In some embodiments, a ratio between thickness **211t2** to thickness **211t1** can range from about 0.1 to about 1.5. If the ratio is less than about 0.1, the tantalum concentration in work function layer **211\*** and gate dielectric layer **203** can be less than about 0.05% and the doped tantalum in dielectric layer **203** may not be able to prevent oxygen diffusing to interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. If the ratio is greater than about 1.5, the tantalum concentration in work function layer **211\*** and gate dielectric layer **203** can be greater than about 25% and the excessive tantalum may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**.

(49) In some embodiments, the formation of work function metal layer **211\*** can be followed by depositing a glue layer on work function metal layer **211\*** to form work function metal layer **211**, as shown in FIG. **16**. In some embodiments, the glue layer can be deposited by the same method and include the same conductive material as the glue layer described in FIG. **12**. In some embodiments, the formation of work function metal layer **211** can be followed by depositing a metal fill **213** on the glue layer, as shown in FIG. **17**. In some embodiments, metal fill **213** can be

deposited by the same method and include the same conductive material as metal fill **213** described in FIG. **13**.

(50) In some embodiments, work function metal layer **211** can include a work function material of TiTaN to dope tantalum in work function metal layer **211** and gate dielectric layer **203**, as shown in FIGS. **18-20**. FIG. **18** illustrates a cross-sectional view of semiconductor device **100** shown in FIG. **8** after operations **520** and **530**, in accordance with some embodiments. FIG. **19** illustrates an enlarged cross-sectional view of region **1870** in FIG. **18**, in accordance with some embodiments. In some embodiments, as shown in FIG. **19**, work function metal layer **211\*** can include a work function material having tantalum, such as TiTaN. In some embodiments, the tantalum concentration in work function metal layer **211\*** can range from about 0.05% to about 25%. If the tantalum concentration is less than about 0.05%, the tantalum in work function metal layer **211\*** and gate dielectric layer **203** may not be able to prevent oxygen from diffusing to interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. As a result, the electrical performance of semiconductor device **100** may not be improved. If the tantalum concentration is greater than about 25%, the excessive tantalum may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**. The decrease of the uniformity of gate dielectric layer **203** can degrade the electrical performance of semiconductor device **100**.

(51) In some embodiments, work function metal layer **211\*** can be deposited on gate dielectric layer **203** by ALD, CVD, and other suitable deposition methods at a temperature from about 150° C. to about 550° C. under a pressure from about 0.1 torr to about 50 torr. In some embodiments, work function metal layer **211\*** can be deposited using a titanium precursor (e.g., TiCl<sub>4</sub>), a tantalum precursor (e.g., PDMAT), and a nitrogen precursor (e.g., ammonia (NH<sub>3</sub>)) to form TiTaN. In some embodiments, work function metal layer **211\*** can be deposited in a chamber by a sequence of the precursors. The sequence of the precursors can include a first cycle of the titanium precursor, a second cycle of the nitrogen precursor, a third cycle of the tantalum precursor, and a fourth cycle of the nitrogen precursor. In some embodiments, the titanium precursor can be delivered to the chamber for a first pulse time during the first cycle and the tantalum precursor can be delivered to the chamber for a second pulse time during the third cycle. In some embodiments, a ratio of first pulse time to the second pulse time can range from about 0.15 to about 10. If the ratio is less than about 0.15, the tantalum concentration in work function layer **211\*** can be less than about 0.05% and the tantalum in work function layer **211\*** may not be able to prevent oxygen diffusion. If the ratio is greater than about 10, the tantalum concentration in work function layer **211\*** can be greater than about 25%. The excessive tantalum may cause defects in gate dielectric layer **203** and decrease the uniformity of gate dielectric layer **203**. In some embodiments, the first cycle can include the titanium precursor but may not include the tantalum precursor, because tantalum deposited on gate dielectric layer **203** may increase V<sub>t</sub> of PFET devices and degrade device performance. In some embodiments, the deposited tantalum can diffuse into gate dielectric layer **203** during the deposition process. The doped tantalum in gate dielectric layer **203** can attract oxygen and mitigate oxygen diffusion. In some embodiments, additional cycles of the titanium precursor, the nitrogen precursor, the tantalum precursor can be delivered to the chamber in the same sequence as the first four cycles to form work function metal layer **211\***. In some embodiments, work function metal layer **211\*** can have a thickness ranging from about 1 nm to about 10 nm.

(52) In some embodiments, the formation of work function metal layer **211\*** can be followed by depositing a glue layer on work function metal layer **211\*** to form work function metal layer **211**, which can be followed by depositing a metal fill **213** on the glue layer, as shown in FIG. **20**. In some embodiments, the glue layer can be deposited by the same method and include the same conductive material as the glue layer described in FIG. **12**. In some embodiments, metal fill **213** can be deposited by the same method and include the same conductive material as metal fill **213** described in FIG. **13**.

(53) In some embodiments, the formation of work function metal layer **211** and metal fill **213** can be followed by a chemical mechanical polishing (CMP) process to planarize top surfaces of gate structure **155**, gate spacers **160**, etch stop layer **145**, and isolation layer **150**. In some embodiments, additional operations can follow the CMP process to form contacts on S/D structures **140**, contacts on gate structure **155**, interconnects, and other structures for semiconductor device **100**, which are not described in detail for simplicity.

(54) Various embodiments of the present disclosure provide methods for forming semiconductor device **100** having a tantalum doped work function metal layer **211** to prevent oxygen diffusion and improve device threshold voltage. In some embodiments, work function metal layer **211** and gate dielectric layer **203** can be doped with tantalum by a soak process with a tantalum precursor. The tantalum in work function metal layer **211** and gate dielectric layer **203** can attract oxygen and prevent oxygen from diffusing into interfacial layer **207** and the interface between interfacial layer **207** and high-k dielectric layer **209**. In some embodiments, work function metal layer **211** can include one or more tantalum nitride layers to dope tantalum in work function metal layer **211** and gate dielectric layer **203** and thus to mitigate oxygen diffusion. In some embodiments, work function metal layer **211** can include TiTa<sub>N</sub> to mitigate oxygen diffusion. The tantalum in the TiTa<sub>N</sub> work function metal layer can have a concentration ranging from about 0.05% to about 25%, and the tantalum can diffuse into gate dielectric layer **203** to attract oxygen and mitigate oxygen diffusion. In some embodiments, the tantalum concentration in work function metal layer **211** and gate dielectric layer **203** can range from about 0.05% to about 25%. In some embodiments, with tantalum doping in work function metal layer **211** and gate dielectric layer **203**, V<sub>t</sub> shift in PFETs **105A** and **105B** can be reduced by about 20 mV to about 100 mV and device performance of semiconductor device **100** can be improved.

(55) In some embodiments, a method includes forming a gate dielectric layer on a channel structure and forming a work function metal layer on the gate dielectric layer. The gate dielectric layer includes an interfacial layer on the channel structure and a high-k dielectric layer on the interfacial layer. The method further includes doping the work function metal layer and the gate dielectric layer with tantalum.

(56) In some embodiments, a method includes forming a gate dielectric layer on a channel structure. The gate dielectric layer includes an interfacial layer on the channel structure and a high-k dielectric layer on the interfacial layer. The method further includes depositing a work function metal layer on the gate dielectric layer with a titanium precursor, a nitrogen precursor, and a tantalum precursor and forming a metal fill on the work function metal layer.

(57) In some embodiments, a semiconductor device includes a channel structure on a substrate, an interfacial layer on the channel structure, a high-k dielectric layer on the interfacial layer, and a work function metal layer on the high-k dielectric layer. The work function metal layer includes tantalum having a concentration ranging from about 0.05% to about 25%.

(58) It is to be appreciated that the Detailed Description section, and not the Abstract of the Disclosure section, is intended to be used to interpret the claims. The Abstract of the Disclosure section may set forth one or more but not all possible embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the subjoined claims in any way.

(59) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

## Claims

1. A method, comprising: forming a gate dielectric layer on a channel structure, wherein the gate dielectric layer comprises an interfacial layer on the channel structure and a high-k dielectric layer on the interfacial layer; forming a work function metal layer on the gate dielectric layer; and diffusing tantalum into the work function metal layer and the gate dielectric layer.
2. The method of claim 1, wherein doping the work function metal layer and the gate dielectric layer comprises soaking the work function metal layer with a tantalum precursor for about 0.5 second to about 1800 seconds.
3. The method of claim 1, wherein doping the work function metal layer and the gate dielectric layer comprises forming a tantalum nitride layer on a titanium nitride layer.
4. The method of claim 3, wherein a ratio of a thickness of the tantalum nitride layer to a thickness of the titanium nitride layer ranges from about 0.1 to about 1.5.
5. The method of claim 1, further comprising: forming an additional work function metal layer on the work function metal layer; and doping the additional work function metal layer with the tantalum.
6. The method of claim 1, further comprising: forming a glue layer on the work function metal layer; and forming a metal fill on the glue layer.
7. The method of claim 1, wherein doping the work function metal layer and the gate dielectric layer comprises performing the doping at a temperature from about 150° C. to about 550° C.
8. The method of claim 1, wherein doping the work function metal layer and the gate dielectric layer comprises performing the doping under a pressure from about 0.1 torr to about 50 torr.
9. The method of claim 1, wherein the work function metal layer is a p-type work function metal comprising titanium nitride, tungsten nitride, tungsten carbon nitride, or titanium silicon nitride.
10. A method, comprising: forming a gate dielectric layer on a channel structure, wherein the gate dielectric layer comprises an interfacial layer on the channel structure and a high-k dielectric layer on the interfacial layer; depositing a work function metal layer on the gate dielectric layer with a titanium precursor, a nitrogen precursor, and a tantalum precursor; diffusing tantalum in the work function metal layer into the gate dielectric layer; and forming a metal fill on the work function metal layer.
11. The method of claim 10, wherein depositing the work function metal layer comprises: delivering the titanium precursor for a first pulse time; and delivering the tantalum precursor for a second pulse time different from the first pulse time.
12. The method of claim 11, wherein a ratio of the first pulse time to the second pulse time ranges from about 0.15 to about 10.
13. The method of claim 10, wherein depositing the work function metal layer comprises performing the deposition at a temperature from about 150° C. to about 550° C.
14. The method of claim 10, wherein depositing the work function metal layer comprises performing the deposition under a pressure from about 0.1 torr to about 50 torr.
15. A method, comprising: forming a gate dielectric layer in a gate stack opening, wherein the gate dielectric layer comprises an interfacial layer and a high-k dielectric layer; forming a work function metal layer on the gate dielectric layer; diffusing tantalum into the work function metal layer and the gate dielectric layer under a thermal condition; and depositing a metal fill in the gate stack opening and on the work function layer.
16. The method of claim 15, wherein forming the work function metal layer comprises soaking the work function metal layer and the high-k dielectric layer with a tantalum precursor for about 0.5 second to about 1800 seconds.
17. The method of claim 16, wherein the work function metal layer and the high-k dielectric layer are soaked at a temperature from about 150° C. to about 550° C.

18. The method of claim 16, wherein the work function metal layer and the high-k dielectric layer are soaked under a pressure from about 0.1 torr to about 50 torr.
19. The method of claim 15, wherein forming the work function metal layer comprises: forming a titanium nitride layer on the high-k dielectric layer; and forming a tantalum nitride layer on the titanium nitride layer.
20. The method of claim 19, wherein a ratio of a thickness of the tantalum nitride layer to a thickness of the titanium nitride layer ranges from about 0.1 to about 1.5.
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